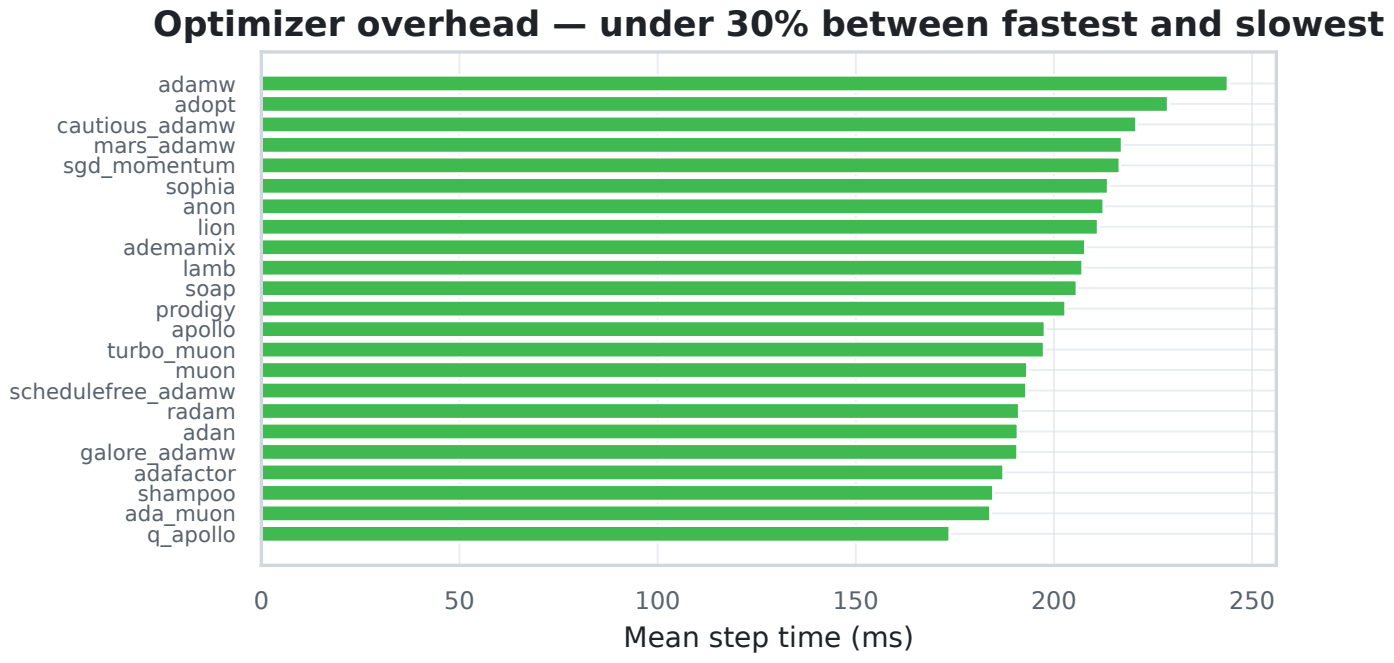
Optimizer overhead — under 30% between fastest and slowest
adamw
adopt
cautious_adamw
mars_adamw
sgd_momentum
sophia
anon
lion
ademamix
lamb
soap
prodigy
apollo
turbo_muon
muon
schedulefree_adamw
radam
adan
galore_adamw
adafactor
shampoo
ada_muon
q_apollo
0
50
100
150
200
250
Mean step time (ms)